Miao Dai

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CONTACT INFORMATION

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POSITION

Assistant Professor, China Economic Research Institute, Liaoning University, Shenyang, China — 2026.09 –

Research Fellow, Vrije Universiteit Amsterdam, the Netherlands — 2026.09 –

Candidate Fellow, Tinbergen Institute, the Netherlands — 2025.03 –

Postdoctoral Researcher, Department of Spatial Economics, School of Business and Economics, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands — 2025.03 – 2026.08

SNSF Project: Critical Minerals and the Clean Energy Transition, with Joëlle Noailly, Steven Poelhekke, and Gerard van der Meijden

Postdoctoral Researcher, University of Montpellier and Center for Environmental Economics – Montpellier (CEE-M), Montpellier, France — 2023.09-2025.02

ANR Project: ScarCyclET – Materials Scarcity and Recycling for the Energy Transition, with Francesco Ricci and Mouez Fodha

EDUCATION

Ph.D. in Economics, McGill University, Montreal, Canada — 2017-2023

Dissertation Title: *Essays in resource market imperfections and their impacts on firm behaviour and the environment*

Committee: Hassan Benchekroun (Supervisor), Francesco Amodio (Co-Supervisor), Laura Lasio, Ngo Van Long.

M.A. in Environmental, Resource and Development Economics (ERDE), University of Winnipeg, Winnipeg, Canada — 2014-2016

B.Sc. in Resources, Environment and Urban and Rural Planning Management, Hubei Normal University, China — 2010-2014

RESEARCH FIELDS

Environmental and Resource Economics, Industrial Organization, International Trade

PUBLICATIONS

"On the profitability of cross-ownership in Cournot nonrenewable resource oligopolies: Stock size matters." with Hassan Benchekroun and Ngo Van Long (2022), *Journal of Environmental Economics and Management*, Volume 111: 102597.

"The impact of cross-ownership on the value of a clean technology in the energy market." with Hassan Benchekroun and Ngo Van Long (2022), *Environmental Modelling and Assessment*, 27(6): 1007-1019.

WORKING PAPERS

"Recycling and strategic extraction in oligopolistic markets for critical minerals", with Hassan Benchekroun, Gerard van der Meijden and Cees Withagen, October 2026.

"Indirect and direct lobbying on international trade in waste", with Guillaume Cheikbossian, October 2026. (Revise and resubmit at *Environmental and Resource Economics*)

"On the impact of cross-ownership in a common property renewable resource oligopoly", with Hassan Benchekroun and Ilyass Dahmouni, September 2026. (Under review)

"On cross-ownership in a nonrenewable resource differentiated product oligopoly", with Hassan Benchekroun, January 2026. (Forthcoming, chapter contribution for *Game theory in economics and management (Essays in honour of Guiomar Martín-Herrán)*)

"Matter matters: Efficient recycling policies under tight markets for scrap", with Francesco Ricci and Mouez Fodha, October 2025. (Draft available upon request)

"Strategic recycling of critical raw materials", with Hassan Benchekroun and Francesco Ricci, October 2025. (Draft available upon request)

"The optimal policy to accelerate electric vehicle adoption: Taxes or subsidies?" with Santiago J. Rubio and Françeska Tomori, October 2025.

"Environmental lobbying on international trade in waste: Theory and evidence", September 2023.

WORK-IN-PROGRESS

"The impact of cross-ownership on the exploitation of a polluting common property renewable resource oligopoly", with Hassan Benchekroun, Gerard van der Meijden and Cees Withagen

"The insiders' dilemma of cross-ownership participants: Experimental evidence", with Hassan Benchekroun, Dimitri Dubois and Marc Willinger

"The pro-competitive effects of cross-ownership: Experimental evidence in a common property renewable resource oligopoly", with Hassan Benchekroun, Dimitri Dubois and Marc Willinger

"Strategic exploitation of a predator-prey ecosystem in the presence of rival cross-shareholdings" with Hassan Benchekroun and Guillaume Bataille

RESEARCH EXPERIENCE

Principal Investigator, "Oligopolistic competition, recycling and strategic trade policy in the geopolitical competition for critical materials in the energy transition" (with Hassan Benchekroun), $10,000 CAD, RRECQ, Canada

Research Assistant

For Hassan Benchekroun, McGill University, 2018-2022
For Amrita Ray Chaudhuri and Soham Baksi, University of Winnipeg, 2015-2017
For Melanie O'Gorman and James Townsend, University of Winnipeg and Public Interest Law Centre, 2016

TEACHING EXPERIENCE

Graduate Teaching Assistant, Max Bell School of Public Policy, McGill University

Global Macroeconomic Policy (Winter 2023, Winter 2022, Winter 2021, Winter 2020)
Microeconomics for Public Policy (Fall 2021, Fall 2020, Fall 2019)

Teaching Assistant, Department of Economics, McGill University

Natural Resource Economics (Winter 2023, Winter 2022)
Advanced Economic Theory - Honours (Fall 2022, Fall 2021)

Microeconomic Analysis and Applications (Summer 2021)
Microeconomic Theory (Winter 2021, Winter 2020, Winter 2018)
Political Economy of Trade Policy (Winter 2021, Winter 2019)
Industrial Organization (Fall 2020)
Public Sector Economics (Fall 2019, Fall 2018)
Mathematics for Economists (Summer 2019)
Economics of the Environment (Fall 2017)

Instructor, Department of Economics, University of Winnipeg

Intermediate Microeconomics (Spring 2017)

Lab Demonstrator/Teaching Assistant, Department of Economics, University of Winnipeg

Intermediate Microeconomics (Winter 2017, Fall 2016, Winter 2016, Fall 2015)
Introduction to Macroeconomics (Winter 2017)
Modern Microeconomics for Economics and Business (Winter 2017)
Mathematical Economics (Fall 2016)
Introduction to Microeconomics (Fall 2016)
Introduction to Economic Theory (Fall 2015)

Seminars and Conferences

2026: CEE-M Workshop on Dynamic Problems in Environmental and Resource Economics, University of Barcelona, Workshop LOBBIES: the Economic Role of LOBBIES, Joint Workshop on Circular Economy Economic Modelling, 7th World Congress of Environmental and Resource Economists (WCERE 2026)

2025: 13th Annual Conference of the Italian Association of Environmental and Resource Economists (IAERE), VU Amsterdam-Department of Spatial Economics Eureka Lunch Seminar, SNSF Critical Minerals Internal Workshop, 30th Annual Conference of the European Association of Environmental and Resource Economists (EAERE), 4th Workshop on Materials Scarcity and Recycling for the Energy Transition, Annual Conference of the European Association for Research in Industrial Economics (EARIE), 5th AERNA Workshop on Game Theory and the Environment, 26th Annual Conference of the European Trade Study Group (ETSG), Nederlandse Economendag (Dutch Economists' Day)

2024: 3rd Workshop on Materials Scarcity and Recycling for the Energy Transition, McGill-CIREQ Lunch Seminar, CIRANO-CIREQ Workshop on Environmental and Natural Resource Economics, Second Young AERNA Day, Paris School of Economics Trade and Environment Transition Workshop, Conference on Sustainable Resource Use and Economic Dynamics (SURED), 23rd Journées Louis-André Gérard-Varet (LAGV), Joint Workshop on Circular Economy Economic Modelling, 29th Annual Conference of the European Association of Environmental and Resource Economists (EAERE), CEE-M Environmental Economics Seminar, Annual meeting of the Association of Southern European Economic Theorists (ASSET)

2023: 28th Annual Conference of the European Association of Environmental and Resource Economists (EAERE), EAERE Summer School – University of Graz, 2023 North America Conference of the Chinese Economists Society (CES)

2022: Beijing Normal University Bay Area International Business School, 32nd Annual Conference of the Canadian Resource and Environmental Economics Association (CREEA), 11th Congress of the Asian Association of Environmental and Resource Economics (AAERE), Xiangjiang River Forum in Economics, Finance and Management for Young Scholars – CEFMS, 12th International Economics and Finance Society (IEFS) China Annual Conference (invited), 2022 Annual Conference of the Chinese Economists Society (CES), 17th Annual Meeting of the Asia Pacific Trade Seminars (APTS), 8th Canadian PhD and Early Career Researcher Workshop in Environmental and Resource Economics, 56th Annual Meeting of the Canadian Economics Association (CEA), 15th Australasian Trade Workshop (ATW), Montreal Environmental and Natural Resources Economics Workshop, McGill-CIREQ Lunch Seminar

2021: Montreal Environmental and Natural Resources Economics Workshop
2020: 30th Annual Conference of Canadian Resource and Environmental Economics Association

Professional Services

Review Editor of Resource Economics (specialty section of *Frontiers in Environmental Economics*)

Referee for *Journal of Environmental Economics and Management*, *Resource and Energy Economics*, *Applied Economics*, *China Economic Review*, *Environmental and Resource Economics*, *Canadian Journal of Economics*, *Environment and Development Economics*

Thesis supervisor (BSc in Economics and Business Economics, VU Amsterdam, 2025)

Fellowships and Awards

STP - Social Sciences and Humanities Research Council of Canada (SSHRC), 2021-2024
STP - Fonds de recherche du Québec - Société et culture (FRQSC), 2018-2024
Participation Grant, Global Excellence in Modeling Climate and Energy Policies (GEOCEP), 2023
Travel Scholarship, European Association of Environmental and Resource Economists, 2023
International Graduate Mobility Award, McGill University, 2023 and 2019
Travel Subsidy, Chinese Economists Society (CES), 2023
Travel Grant, Canadian Resource and Environmental Economics Association (CREEA), 2022
Lasserre-Renzetti Prize for Best Student Paper, CREEA, 2022
Student Travel Grant, Canadian Economics Association (CEA), 2022
Graduate Research Enhancement and Travel Award, Faculty of Arts, McGill University, 2022
Clifford C F Wong Fellowship - Economics, McGill University, 2017-2020
14th Competition and Regulation European Summer School (CRESSE) Fellowship, 2019
Graduate Excellence Award in Economics, McGill University, 2017

Professional Affiliations

Québec Circular Economy Research Network/Réseau de recherche en économie circulaire du Québec (RRECQ)
Canadian Resource and Environmental Economics Association (CREEA)
European Association of Environmental and Resource Economists (EAERE)
French Association of Environmental and Resource Economists (FAERE)
Spanish-Portuguese Association of Natural and Environmental Resource Economics (AERNA)
Italian Association of Environmental and Resource Economists (IAERE)
Chinese Economist Society (CES)

Languages

Chinese (native), English (fluent), French (basic)

Skills

Matlab, Mathematica, Stata, LaTeX